Ignytion io

IGNYTION IO · TECHNICAL LAB NOTE

UART RTL Co-Simulation with Renode and Verilator

Build, run, check, and view a SpinalHDL APB3 UART through a Makefile

Renode drives UART RTL through a Direct Programming Interface (DPI). A Makefile calls the build and smoke scripts and records the same execution as a Value Change Dump (VCD). The measured checks cover register values, serial loopback, and interrupt states. Crucible beta supplies the pinned tools.

#	WHAT WE RAN	RESULT
1	9 register values and 5 interrupt states	✓ PASS
2	Real model accepted; two controls rejected	✓ PASS
3	Same-run VCD: 387 signal declarations	✓ PASS

SECTION 01

What this note establishes

A UART can be checked at its software-visible registers while its serial transmitter, receiver, and interrupt logic execute in RTL. This note provides a Makefile workflow for that co-simulation. Its build, checked run, and waveform targets use the same DPI model.

The device is a **SpinalHDL APB3 UART**. **Renode 1.17.0** drives its bus, and **Verilator 5.052** executes the generated model. TX is wired to RX; the interrupt pin returns through Renode GPIO. **Crucible beta** pins the tools in `crucible.lock`.

THE THREE THINGS IT SHOWS

CLAIM	EVIDENCE	STATUS
Makefile build and checked run	Crucible-managed Make	✓ PASS
UART data and interrupt behaviour	14 typed RTL readbacks match	✓ PASS
Waveform of the same execution	VCD bus, serial and IRQ checks	✓ PASS

> **Measured functional scope**
> Validated on **Linux x86_64 under WSL 2**. Scope is the supplied register, serial-loopback, and IRQ sequence. Cycle accuracy, silicon timing, and complete UART compliance remain untested.

SECTION 02

The system under test

Renode and the Verilated UART run as separate processes. One local TCP connection carries bus requests and replies; a second carries asynchronous events, including interrupts.

Verilator translates the UART RTL into C++. **CMake** configures compilation of that code, the Renode DPI bridge, and its socket library. The **C++ compiler and linker** produce `output/apb3uart/verilated`. Renode launches that executable and converts register accesses into real **APB3 transactions**.

TX is wired to RX in the simulation top, and the **IRQ pin** returns through Renode GPIO. UART RTL is fetched from pinned Antmicro sources under MIT. Adapted DPI glue retains Apache-2.0 attribution.

THE REGISTERS THE RUN TOUCHES

OFFSET	REGISTER	FIELDS THE RUN USES
`0x00`	DATA	write: TX byte; read: RX valid [16], byte [7:0]
`0x04`	STATUS	RX count [31:24], TX free [23:16], IRQs and enables
`0x08`	CLOCK_DIVIDER	baud divider, set to 0
`0x0C`	FRAME	8 data bits, no parity, one stop bit

SECTION 03

Driving it from a Makefile

The Makefile calls the project's build and smoke scripts through the Igny CLI. It selects the lock's default environment explicitly. The `sed` target converts shell/configuration files to Linux line endings and also runs before build, run, and smoke.

`igny run make build` produces a model with VCD support. **`igny run make run`** executes all readbacks and rejection controls while recording the waveform. **`igny run make smoke`** repeats the checks without recording. **`igny run make waves`** opens the existing VCD in GTKWave; **`igny run make clean`** removes generated output.

```
# run from the project root after workspace setup
igny run make build
[100%] Built target verilated
igny run make run
ok    rtl-model: pass
ok    missing-model: reject
ok    wrong-expected-byte: reject
co-simulation smoke passed
PASS waveform: UART A/Z readbacks, clock, serial line and IRQ activity
```

The run uses **14 typed readbacks** and a completion marker. The checker also rejects Renode errors, operation timeouts, and model assertions. The visible FAIL output in each deliberate negative case is an expected rejection.

SECTION 04

What the co-simulation returns

The checked sequence reads **9 register values and 5 interrupt states**. After transmitting A and Z, Renode advances virtual time so the UART can shift and receive each byte. The early A read also checks that the byte has not arrived immediately.

FOURTEEN READBACKS

READS 1–7	VALUE	READS 8–14	VALUE
STATUS reset	`0x00200000`	DATA rx empty	`0x00000000`
IRQ reset	`False`	DATA 'A' still on the wire	`0x00000000`
STATUS tx+rx irq enabled	`0x00200103`	DATA rx 'A'	`0x00010041`
IRQ tx empty	`True`	STATUS 'Z' waiting	`0x01200202`
STATUS rx irq enabled	`0x00200002`	IRQ rx pending	`True`
IRQ nothing received	`False`	DATA rx 'Z'	`0x0001005A`
STATUS irqs disabled	`0x00200000`	IRQ rx drained	`False`

THE SMOKE HAS TWO CONTROLS THAT MUST FAIL

CASE	EXPECTED	WHY IT MATTERS
The real model	pass	14 values and completion match
A missing model	reject	missing executable causes startup failure
A wrong expected byte	reject	actual Z rejected against a wrong byte

SECTION 05

The waveform of the same run

igny run make run sets a trace-file path for the Verilated executable. Its main loop records model activity during the checked DPI execution and closes the VCD when Renode disconnects. The waveform checker reads the resulting file.

```
# validate the VCD produced by igny run make run
python3 scripts/check_waveform.py output/waves/cosim.vcd
PASS waveform: UART A/Z readbacks, clock, serial line and IRQ activity
signals: 387 declarations, 238 unique codes
samples: 500004 timestamps; size: 15551251 bytes
igny run make waves
GTKWave Analyzer v3.4.0
```

The recorded **read-data bus** contains `0x00010041` and `0x0001005A`. The **clock**, **serial loopback wire**, and **IRQ** each show low and high states. The 387 declarations share 238 signal codes because aliases refer to the same recorded signal. A truncated trace and a trace with the A readback corrupted were both rejected.

SECTION 06

Reproduce the Makefile workflow

PROCEDURE

```
igny init
igny workspace sync
igny tool install make --env default
igny run make build
igny run make run
igny run make smoke
# optional graphical waveform viewer
igny tool install gtkwave --env default
igny run make waves
# remove generated output when finished
igny run make clean
```

igny run make sed is available separately; build, run, and smoke already prepare Linux line endings. The build requires **CMake 3.19+**, Make, a C++ compiler, Perl, Python 3, Git, and timeout on PATH. The measured run used **Igny 1.0.0**, CMake 4.2.3, and the locked tool pair.

Limits and cleanup

Native Windows and macOS, cycle accuracy, baud-rate coverage, parity/framing faults, and FIFO stress are untested. GTKWave needs a graphical session. Clean removes output only and refuses an output symlink; a later build fetches and compiles again.

SOURCE NOTES

UART RTL: Antmicro, MIT, commit c9692ca3; tool versions pinned by crucible.lock
`github.com/antmicro/renode-verilator-integration`

DPI source adaptations: Antmicro, Apache-2.0; Renode 1.17.0 and Verilator 5.052
`github.com/antmicro/renode-dpi-examples`